E4.05 - The Answers 2023

Q1 a)

i) The probability that i customers arrive in m subintervals is a binomial distribution

$$P(X=i) = \binom{m}{i} [\lambda h + o(h)]^i [1 - \lambda h + o(h)]^{m-i}$$

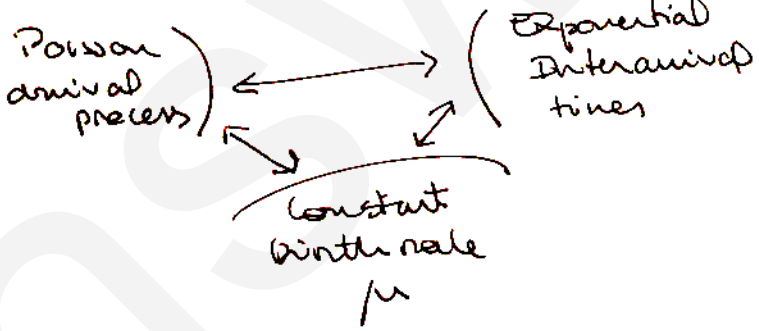
Taking the limit as $h = \Delta t \to 0$ and $m \to \infty$

$$P(X=i) = \frac{(\lambda h)^i}{i!} e^{-\lambda h}$$

[4]

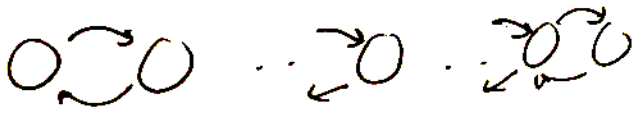
ii) constant birth rate $= \mu$

The relationships



[3]

iii) For a 1-D the matrix G is block diagonal allowing only transitions among adjacent states. Eg the birth model



[3]

Q1a)

iv) 30 calls/m, 0.5 m $\Big)$ $\rho = 30 \times 0.5 = 15$ erlangs

$N = 24$ from Traffic/Erlang chart

we obtain $B_c \sim 1.0\%$ [3]

Q1b)

$0 < i \leq 2$

$i\mu\pi_i = [(M_1 - i + 1)\lambda_1 + (M_2 - i + 1)\lambda_2]\pi_{i-1}$

i) $M_1 = 200$ $M_2 = 10$ $\frac{1}{\mu} = 120\,s$ $(0.0\bar{3}\,h)$

$\lambda_1 = 0.3/h$ $\lambda_2 = 6/h$

$\frac{\lambda_1}{\mu} = 0.01$ $\frac{\lambda_2}{\mu} = 0.2$ [4]

ii)

$i=1)$ $\pi_1 = [(200)0.01 + [10)0.2] \cdot \pi_0$

$\pi_1 = [2+2]\pi_0 \longrightarrow \boxed{\pi_1 = 4\pi_0}$

$i=2)$ $2\pi_2 = [201-2)0.02 + (11-2)0.2] \cdot \pi_1$

$[199 \times 0.02 + 9 \times 0.2]\pi_1$

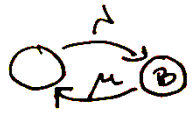
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$\boxed{\pi_2 = 2.89\pi_1}$

$\boxed{\pi_1 + \pi_2 + \pi_0 = 1}$ [3]

Answers

Q2a)

i) 

$P_{\text{source busy}} = \frac{\lambda}{\lambda+\mu} = P$

If M (number of sources) $> N$ (number of channels)

$$\pi_i = \frac{\binom{M}{i} p^i (1-p)^{M-i}}{\sum_{j=0}^{N} \binom{M}{j} p^j (1-p)^{M-j}} \qquad i = 0, 1, \ldots N$$

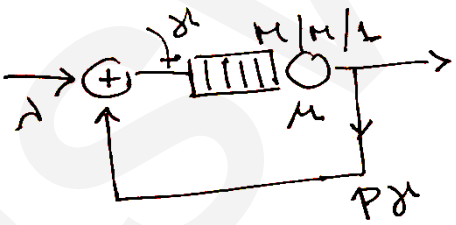
[5]

ii) $p = 0.01 \quad M = 5 \quad N = 2$

$$\pi_2 = \frac{\binom{5}{2}(0.01)^2 (0.99)^3}{\binom{5}{0} p^0 (0.99)^5 + \binom{5}{1}(0.01)^1 (0.99)^4 + \binom{5}{2}(0.01)^2 (0.99)^3}$$

[5]

Q2b)

i) 

$\gamma = \lambda + p\gamma$

$\gamma = \frac{\lambda}{1-p} \qquad \rho = \frac{\gamma}{\mu}$

$$P(Q_t = i \mid N_t \geq 1) = \frac{P(N_t = 1+i)}{\sum_{j=0}^{\infty} P(N_t = 1+j)} = (1-\rho)\rho^i$$

[6]

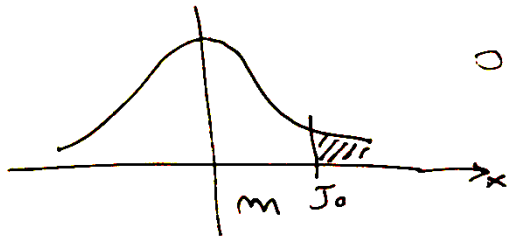
ii) $E[Q_t \mid \text{Delay}] = \frac{\rho}{1-\rho}$

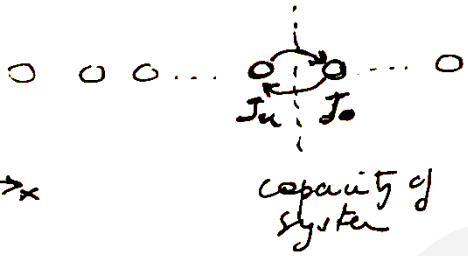
[4]

Answers

Q3a)

i)





this function is an approximation of a binomial distribution (right hand side) by a Gaussian distribution (left hand side)

ε is an approximate to the probability of being in the overload region of the multiplexer. [5]

ii) X is a continuous random variable that represents the buffer occupancy.

m = mean number of sources on.

J_0 = the joint overload state [4]

Answers

Q3 b)

Link capacity = 2 Mbits/s = 2×10^6 bits/s

Message = 20 bits

λ = 1000 m/s = 10^3 m/s

i) $E(S) = \frac{20}{2 \times 10^6} \left(\frac{1+2+3+4+5}{5} \right) = \frac{10}{10^6} \times 3 = \frac{30}{10^6}$

$= 30 \mu s$

$\rho = \lambda E(S) = 1000 \times 30 \times 10^{-6} = 0.03$ [4]

ii) $E(S^2) = \left(\frac{20}{2 \times 10^6} \right)^2 \left(\frac{1^2+2^2+3^2+4^2+5^2}{5} \right) =$

$= 1 \times 10^{-10} \left(11 \right) = 1.1 \times 10^{-9} s^2$ [4]

iii) Mean waiting time = $\frac{\lambda E(S^2)}{2(1-\rho)}$ [3]

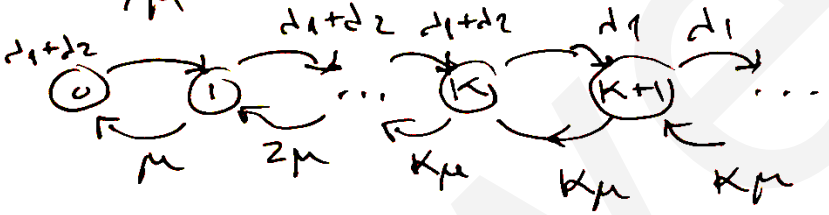
Answers

Q4a) This is a 1D B/D process as the two arrival streams have the same service time distribution

i) - Birth coeff $\lambda_i = (\lambda_1 + \lambda_2) \quad i \leq K$
$= \lambda_1 \quad i > K$

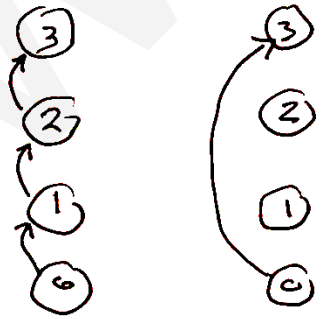
- Death coeff $\mu_i = i\mu \quad i \leq K$
$= K\mu \quad i > K$

$h = 1/\mu$ [2]

ii) 

[4]

Q4b) i) 

ii) 

and any combination

[3] , [5]

iii) 

iv) $R = [R_0, R_1, R_2, R_3]$

R_0 = fault state

Availability reward structure

$R_A = [0, 1, 1, 1]$

[3] , [3]

Answers